



# Advances Toward High Gradient $\text{Nb}_3\text{Sn}$ Thin Film Superconducting Radiofrequency Cavities

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In partnership with:

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Northwestern

# Acknowledgements

- Advisors:
  - David Seidman
  - Sam Posen
- Additional Contributors:

David Burk, Brad Tennis, Chad Thompson, Alexandr Netepenکو, Alex Melnychuk, Robert Schuessler, Damon Bice, Jaeyel Lee, Grigory Ereমেেে

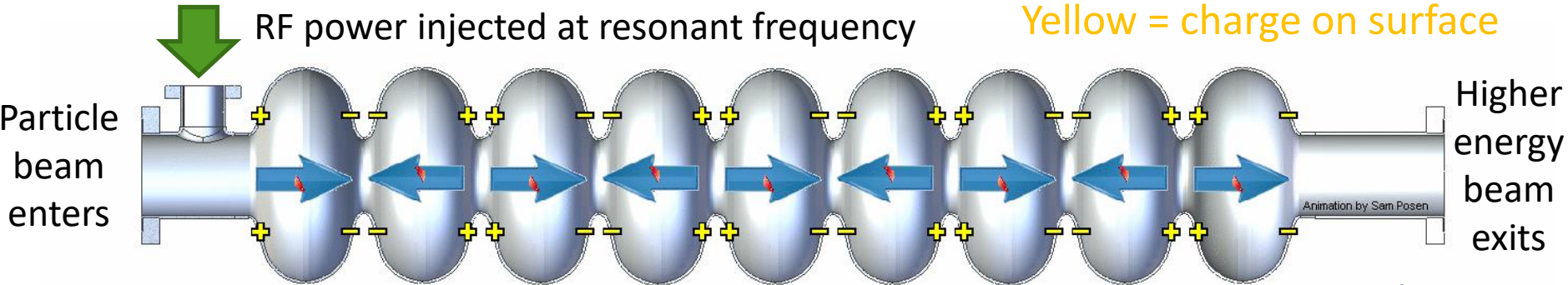
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# Superconducting Radiofrequency (SRF) Cavities

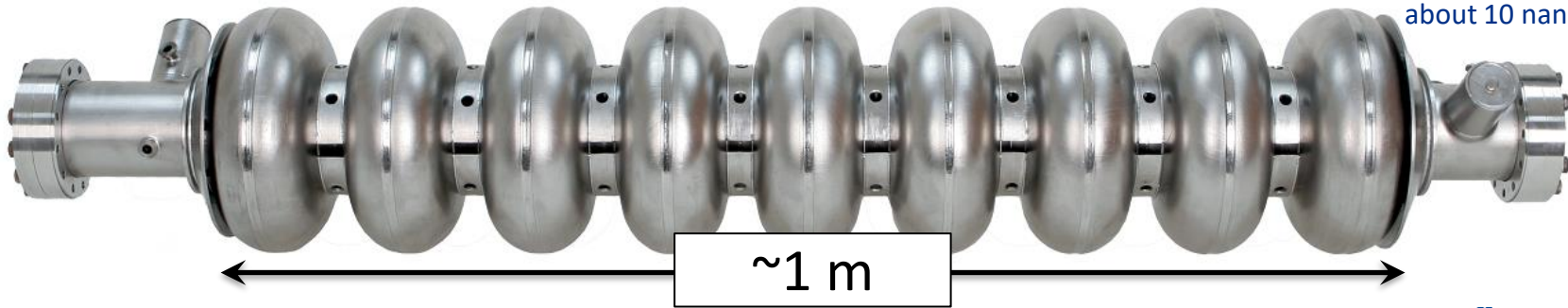
Blue = electric field

Red = magnetic field

Yellow = charge on surface

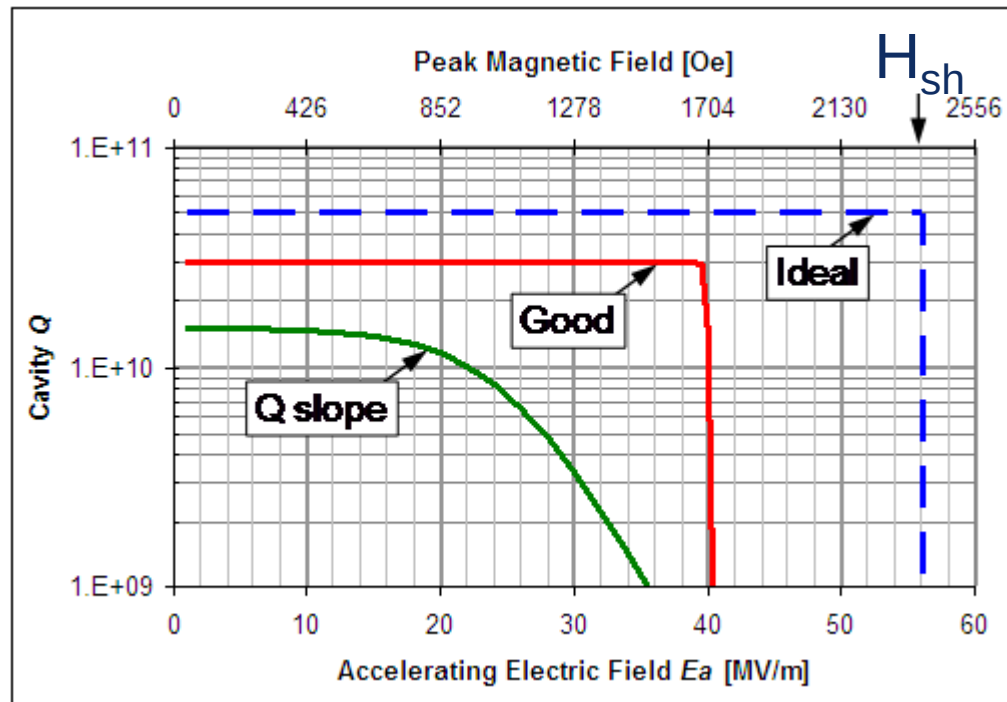


Animation loop represents about 10 nanoseconds



# Quantifying SRF Cavity Performance

- Two major performance metrics:
  - Quality Factor
    - $Q = \frac{\text{Total Energy}}{\text{Surface Resistance}}$
  - Accelerating Field
    - Limited by superheating field  $H_{sh}$ .



# Beyond Niobium – Nb<sub>3</sub>Sn Thin Film Cavities

|                  | Nb               | Nb <sub>3</sub> Sn | Difference |
|------------------|------------------|--------------------|------------|
| Q <sub>@4K</sub> | ~10 <sup>9</sup> |                    |            |
| E <sub>max</sub> | 50 MV/m          |                    |            |

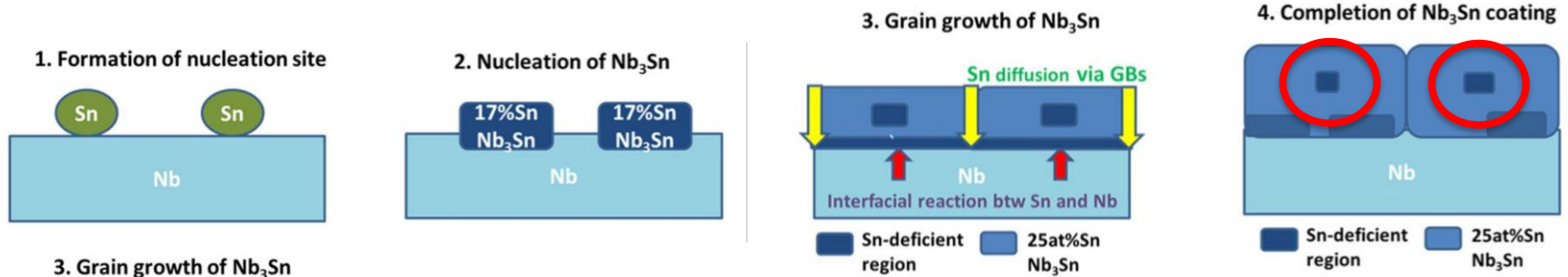
# Beyond Niobium – Nb<sub>3</sub>Sn Thin Film Cavities

|                  | Nb               | Nb <sub>3</sub> Sn | Difference |
|------------------|------------------|--------------------|------------|
| Q <sub>@4K</sub> | ~10 <sup>9</sup> | ~10 <sup>10</sup>  | 10x        |
| E <sub>max</sub> | 50 MV/m          | 100 MV/m           | 2x         |

# Beyond Niobium – Nb<sub>3</sub>Sn Thin Film Cavities

|                  | Nb               | Nb <sub>3</sub> Sn | Difference |
|------------------|------------------|--------------------|------------|
| Q <sub>@4K</sub> | ~10 <sup>9</sup> | ~10 <sup>10</sup>  | 10x        |
| E <sub>max</sub> | 50 MV/m          | ~20 MV/m           | 2x         |

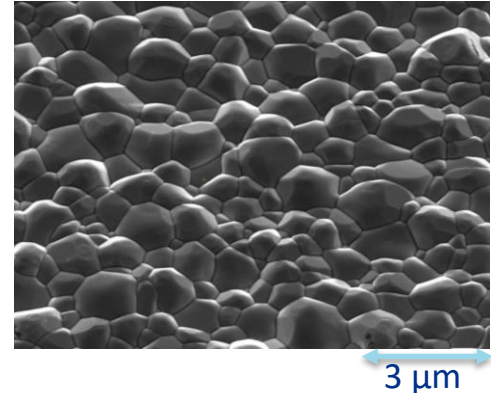
# Challenges of Nb<sub>3</sub>Sn Cavity Fabrication



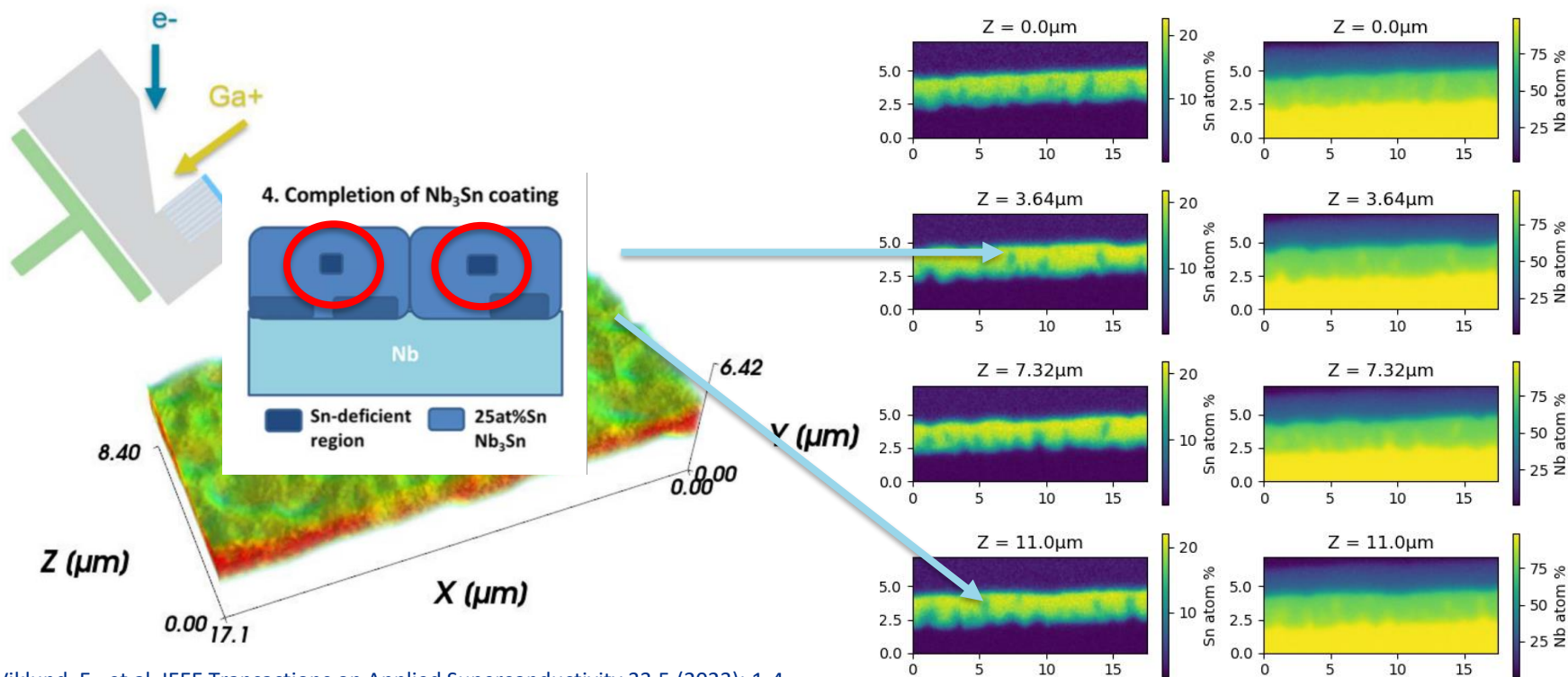
Lee, Jaeyel, et al. Superconductor Science and Technology 32.2 (2018): 024001.

Viklund, Eric, et al. Superconductor Science and Technology 37.2 (2024): 025009.

- Nb<sub>3</sub>Sn is brittle.
- Stoichiometry is important.
- Nb<sub>3</sub>Sn cannot be electropolished.



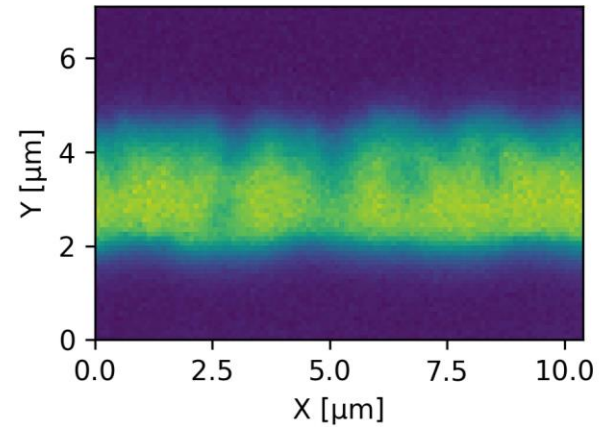
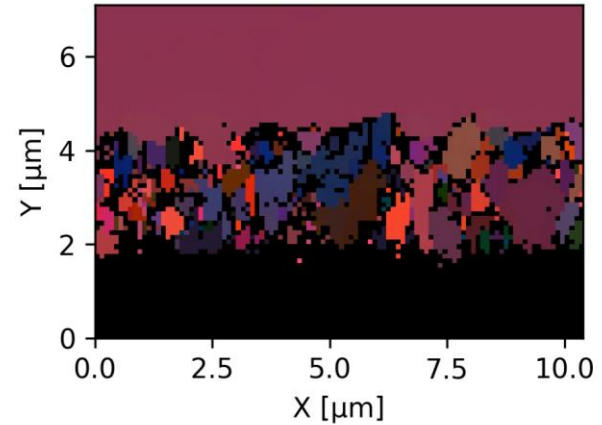
# Investigation of Sn Deficient Regions Using 3D Tomography



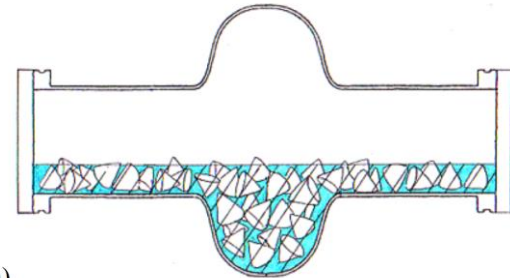
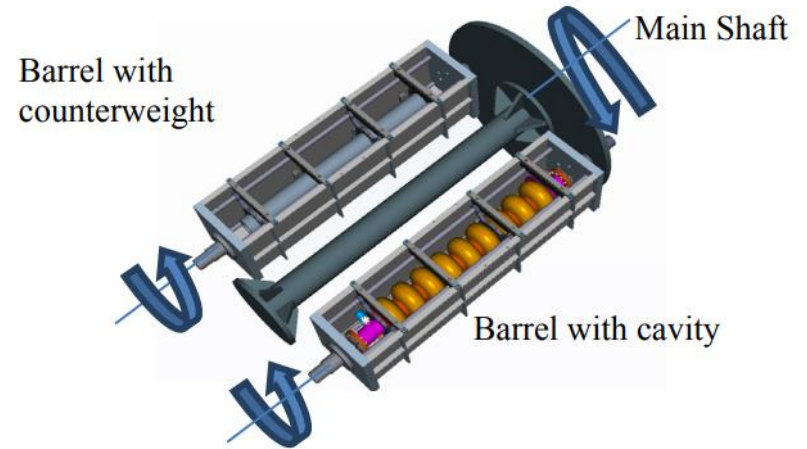
Viklund, E., et al. IEEE Transactions on Applied Superconductivity 33.5 (2023): 1-4.

# Combined 3D EDS and EBSD

- EDS shows Sn distribution within the film.
- EBSD shows the grain structure of the film.
- Sn deficient are shown to be intra granular as suggested by the Sn nucleation model.



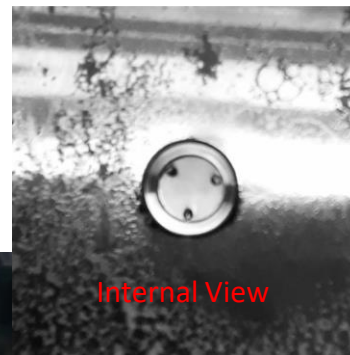
# Centrifugal Barrel Polishing (CBP)

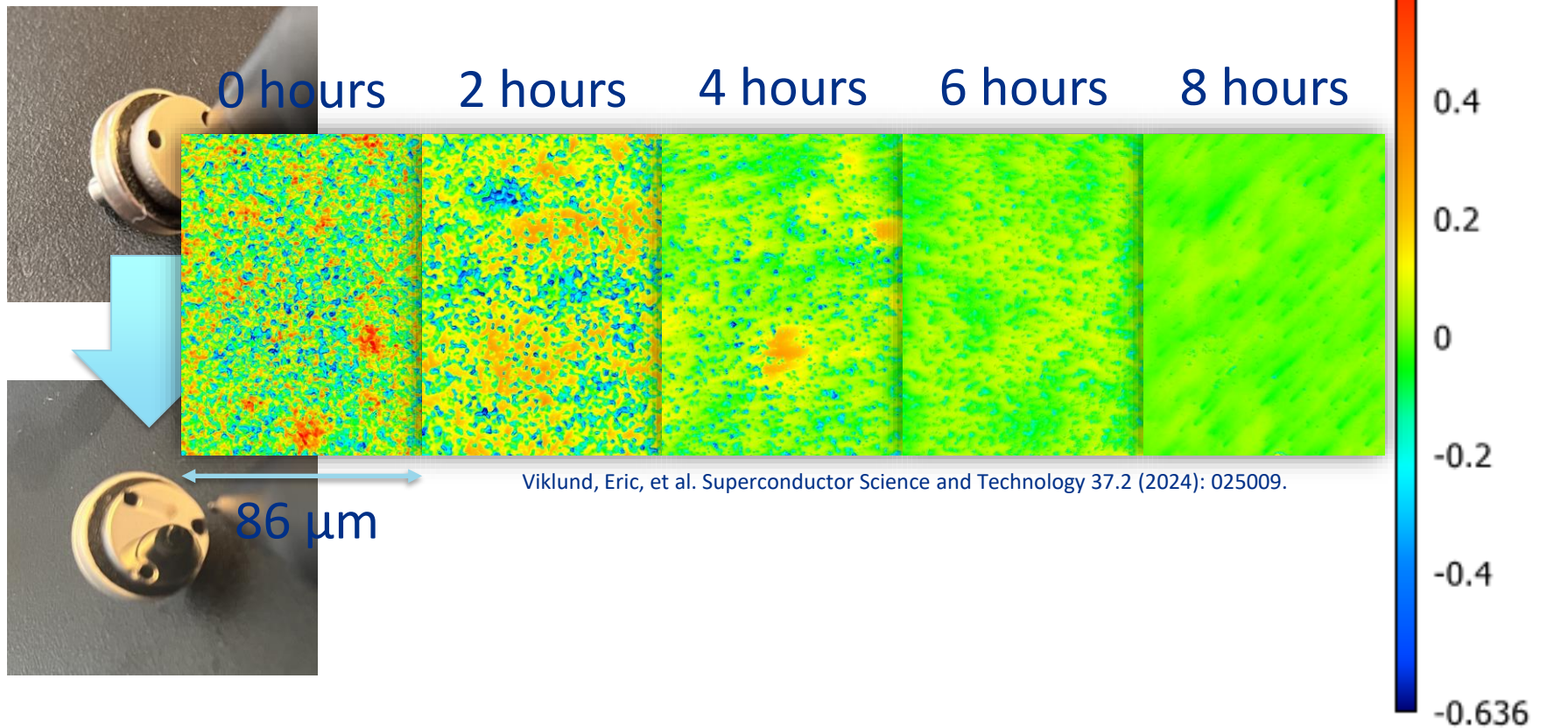


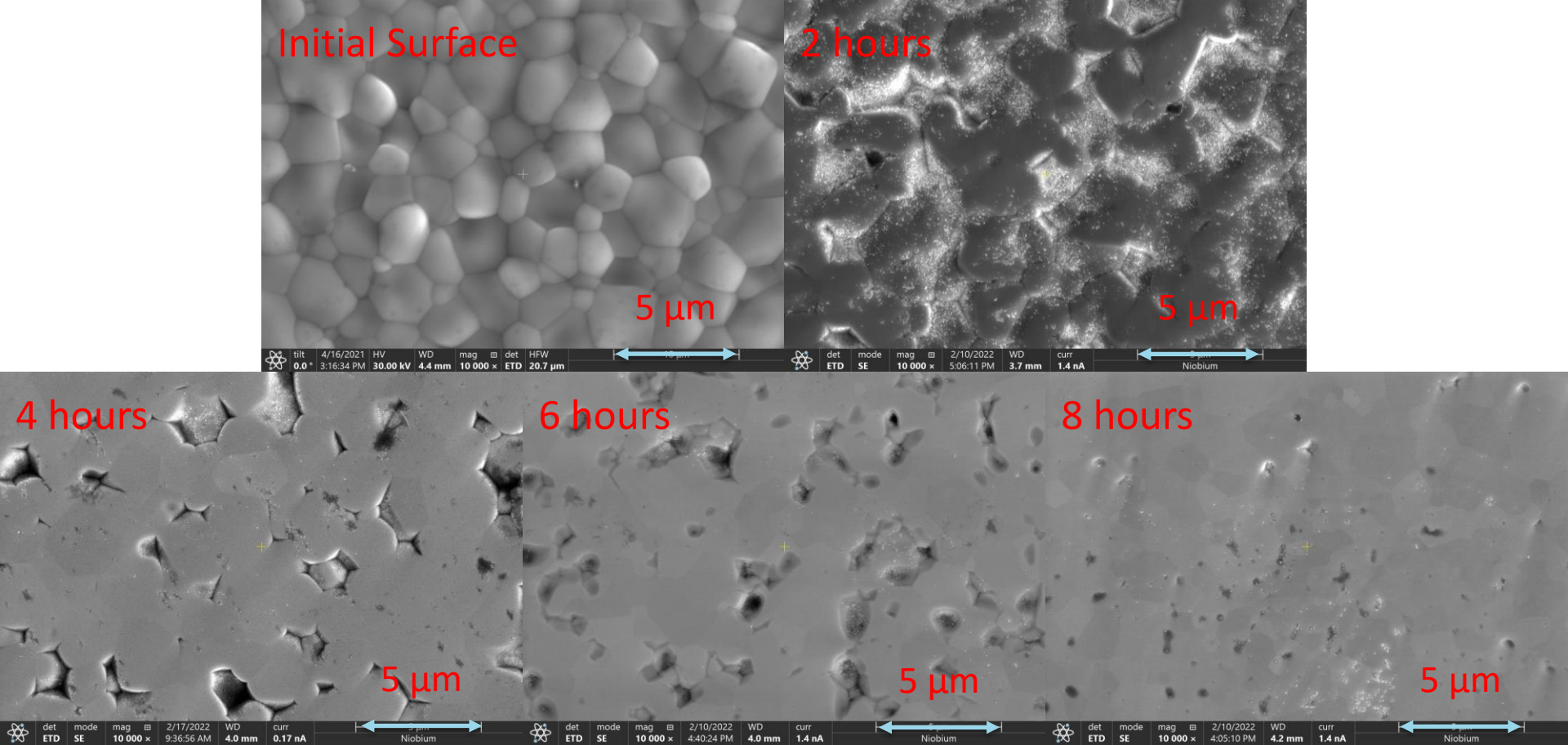
a)

# $\text{Nb}_3\text{Sn}$ Sample Polishing Using a Coupon Cavity

1 cm  $\text{Nb}_3\text{Sn}$   
coated sample

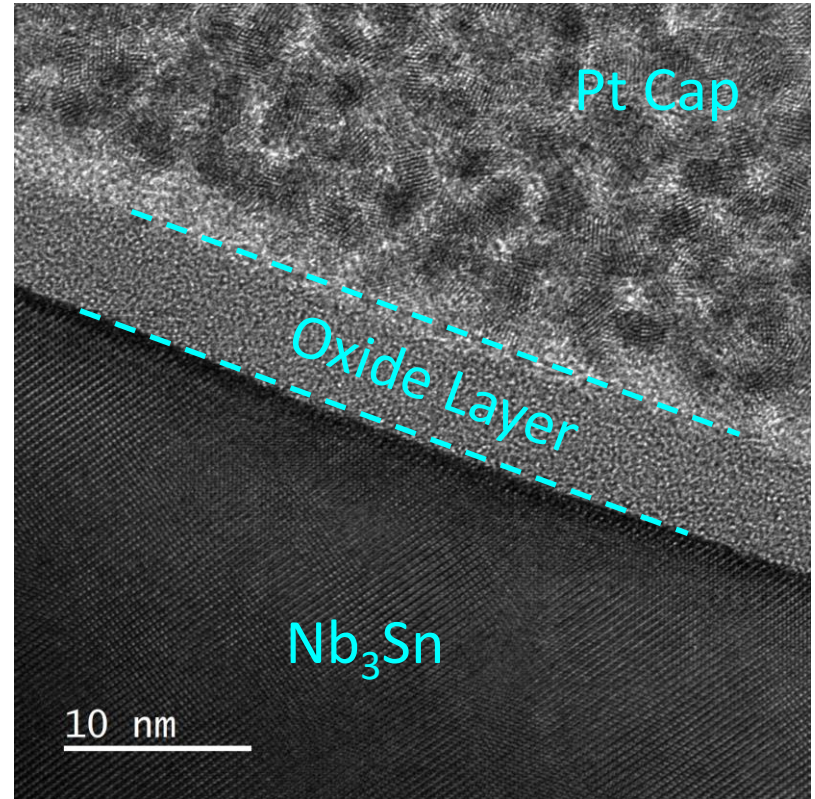
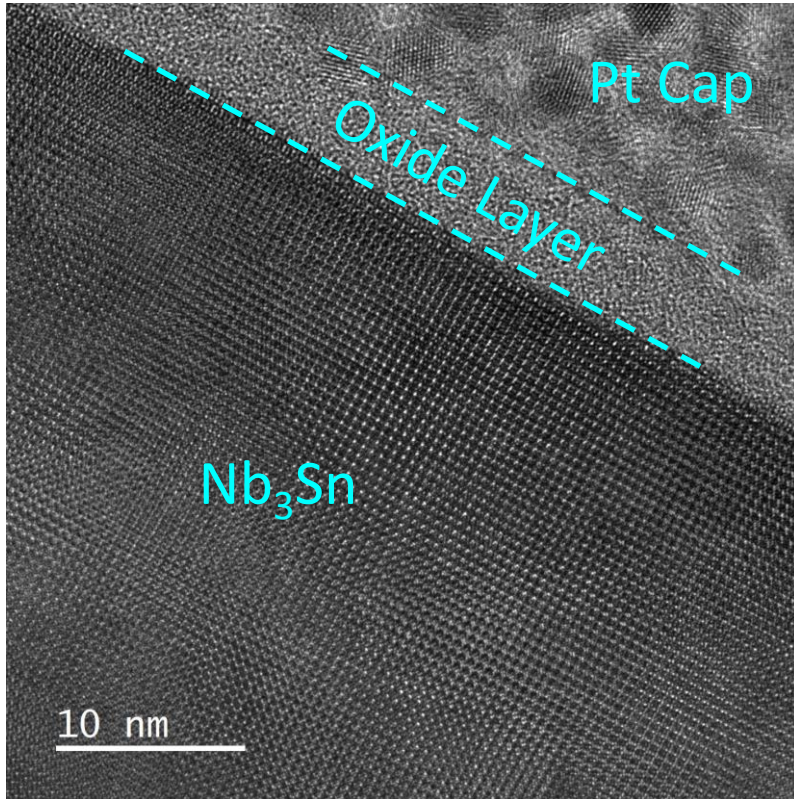




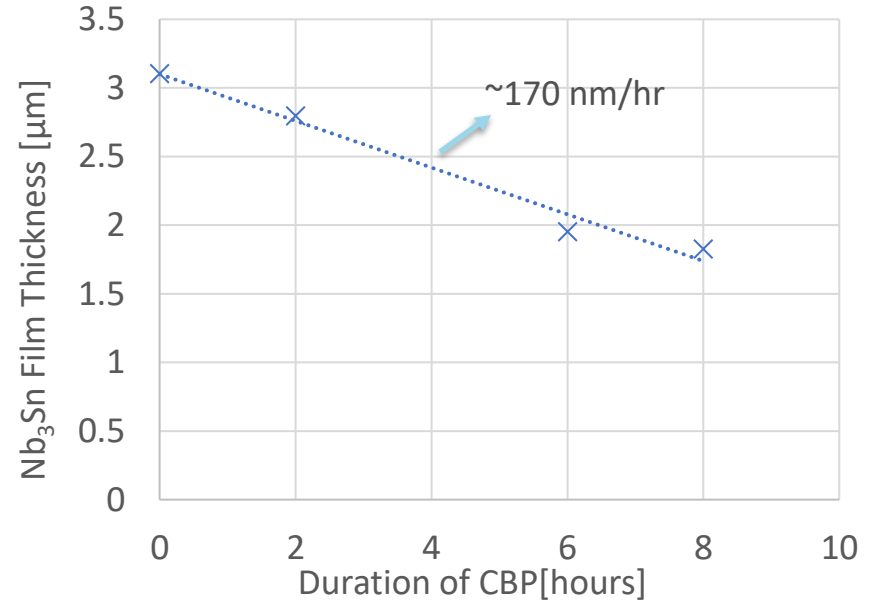
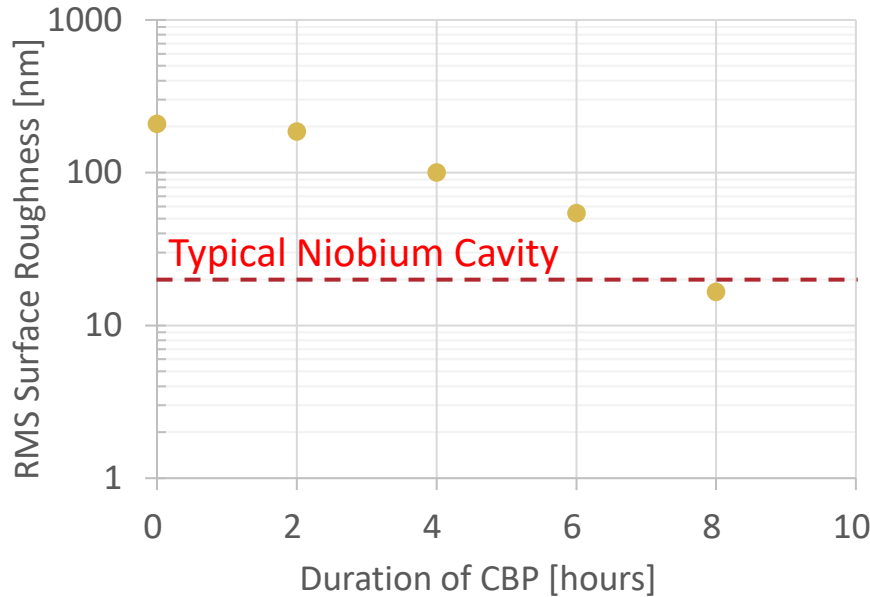


Viklund, Eric, et al. Superconductor Science and Technology 37.2 (2024): 025009.

# Nanoscale Image of a Polished $\text{Nb}_3\text{Sn}$ Thin Film



# CBP Surface Roughness and Removal Rate



Viklund, Eric, et al. Superconductor Science and Technology 37.2 (2024): 025009.

# Applying Centrifugal Barrel Polishing (CBP) to a Nb<sub>3</sub>Sn Cavity

Before Nb<sub>3</sub>Sn Coating



Nb<sub>3</sub>Sn

After Nb<sub>3</sub>Sn Coating



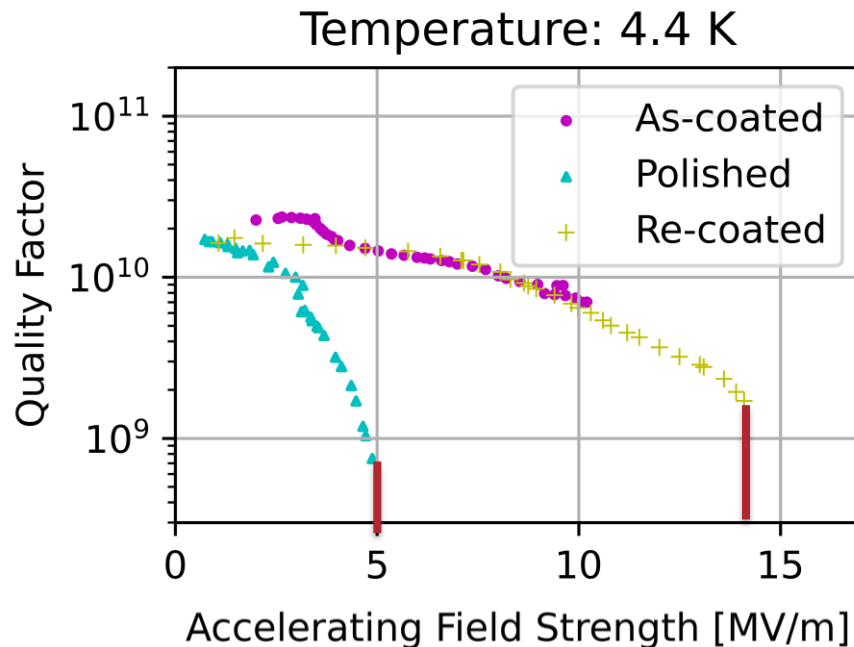
4-hour  
CBP

After Polishing



# Performance of a Mechanically Polished Nb<sub>3</sub>Sn SRF Cavity

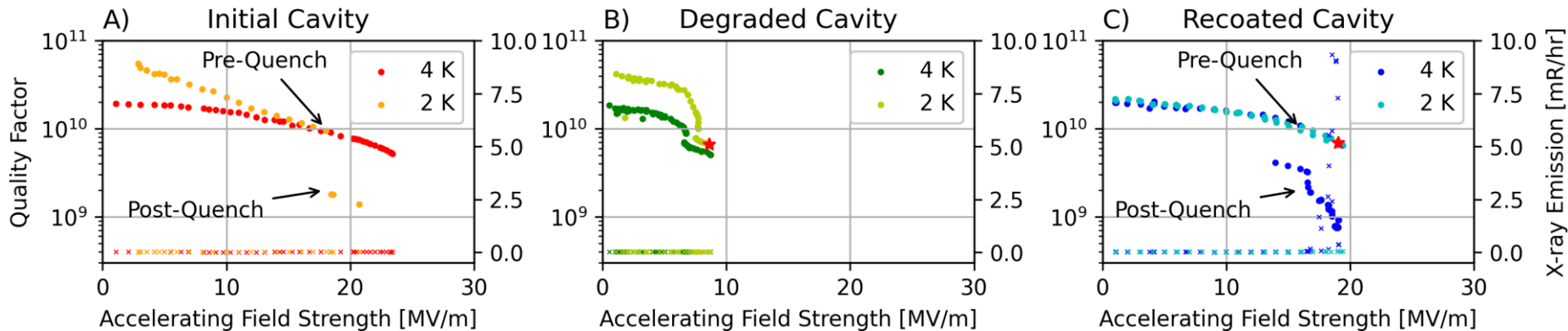
- Polishing initially causes degradation.
- A second coating is applied to fix the degradation.
- Max accelerating field increases.
  - 10 MV/m to 14 MV/m
- Quality factor stays the same.
- First successful application of polishing to a Nb<sub>3</sub>Sn cavity!



Viklund, Eric, et al. Superconductor Science and Technology 37.2 (2024): 025009.

# Recoating Heals Degraded Nb<sub>3</sub>Sn Cavities

- Nb<sub>3</sub>Sn cavity performance is degraded by strain.
  - Tuning
  - Assembly
  - Transportation
- A second, light coating is required to fix performance.
  - 1200 °C → 1000 °C
  - 5 h → 1 h
  - 4 g of Sn → 0.85 g of Sn



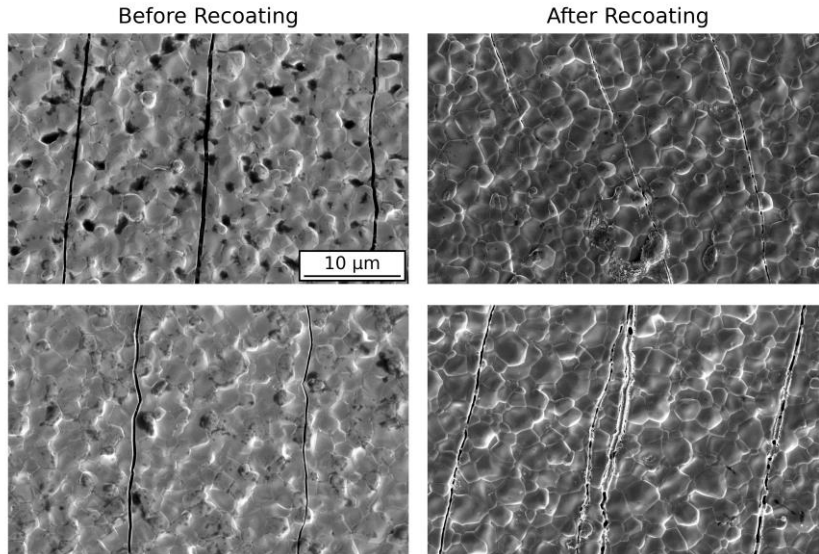
# Finding the Mechanism of Nb<sub>3</sub>Sn Cavity Healing

- Strain applied to cavity causes cracks in Nb<sub>3</sub>Sn coating.
- Somehow the recoating can repair cracks.
  - Are the cracks being filled in?
  - Is the Nb<sub>3</sub>Sn coating re-dissolving?
  - Is the new Nb<sub>3</sub>Sn stoichiometric?
- Nb<sub>3</sub>Sn coated wires are used to study crack formation and healing.
  - 0.1% to 0.5% elongation of the wire causes crack formation.

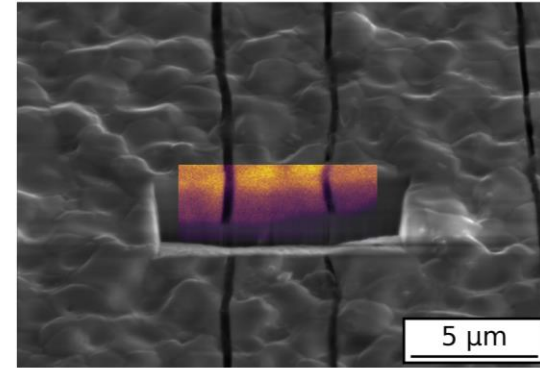


# Results of Wire Study

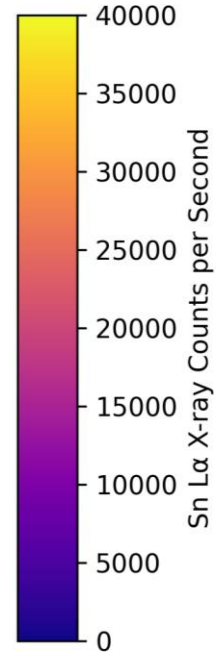
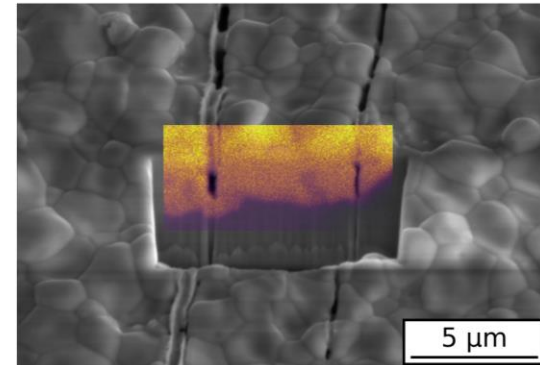
- New  $\text{Nb}_3\text{Sn}$  is formed in the crack and at the base of the crack in the Nb substrate.
- New material is stoichiometric.



0.5% Elongation Before Recoating



0.5% Elongation After Recoating



# Summary

- Through our improved understanding of the  $\text{Nb}_3\text{Sn}$  microstructure, we can implement new manufacturing techniques and achieve higher SRF cavity performance.
- Publications:
  - Viklund, E., et al. IEEE Transactions on Applied Superconductivity 33.5 (2023): 1-4.
  - Viklund, Eric, et al. Superconductor Science and Technology 37.2 (2024): 025009.
  - Viklund, Eric, et al. APL Materials 12.7 (2024).

